

codebase EN script

To resonate with the mentors of the Avalanche Codebase, this document outlines in detail the precedents and challenges related to privacy wallets and their societal implementation, as well as the uniqueness and impact of the solutions offered by GhostWallet.

Please use this as a reference when creating the pitch materials together with the prototype!

- Current Status and Issues of Precedents such as Tornado Cash
- Challenges in MEV, Sandwich Attacks, and Cross-Chain Operations
- Novelty and Architecture of GhostWallet (eERC, G-chain, ZK, DEX Mechanism)
- Regulatory Compatibility (Privacy via KYC Triggers) and Social Implementation Scenarios
- Architecture Designed to Directly Address Investor Needs and Real-World Demand

Current Issues (Privacy / Cross-Chain Domain)

1. Limitations of Existing Privacy Wallets

- **Tornado Cash:** High anonymity, but banned due to regulations
- **Aztec Connect:** Technically advanced, but discontinued (2024)
- **Railgun and others:** Poor UX, slow speed, EVM-centric

2. Vulnerabilities in Cross-Chain DEXs

- Unfavorable execution caused by front-running and sandwich (MEV) attacks
- Swap histories are traceable, leading to privacy leakage
- Complex UX (requires multiple wallets and bridge operations)

GhostWallet's Solution

1. **Complete anonymization and auditability through eERC × G-chain × ZK**
→ Legally compliant privacy via KYC-triggered selective disclosure
2. **Built-in Cross-Chain DEX concealing MEV attacks**
→ Wallet-integrated seamless swapping experience via G-host DEX
3. **Scalable deployment leveraging Avalanche's speed and subnet architecture**

Why GhostWallet Is Uniquely Positioned

- The **only** architecture that combines *legality* and *anonymity* (eERC × KYC Trigger × ZK)
- Provides *cross-chain*, *privacy*, and *MEV resistance* all in one (Other products offer fragmented, single-functionality solutions)
- Highly compatible with the Avalanche ecosystem (AvaCloud, Warp Messaging)
→ A fertile ground for scaling within the Codebase program

👉 Thus, GhostWallet is not just a “privacy wallet” —

It is the **next-generation Web3 standard**, built to withstand MEV, regulatory scrutiny, and UX challenges.

Product Vision

► Problem / Desire:

“I want a wallet that protects privacy while complying with regulations — one I can use with peace of mind.”

(= Solve the dilemma of *anonymity vs. compliance*)

► Global Target Audience:

- Large-scale DeFi users
 - Traders seeking protection from sandwich/MEV attacks
 - Institutional investors and enterprises (e.g. RWA, B2B settlement)
 - Tornado Cash users seeking alternatives / Aztec Connect migrants
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Market Size & Numerical Backing

1. Large DeFi Users (Whales)

- **Target Segment:** Active wallets managing over \$100,000 in DeFi
- **Stats:**
 - ~8 million DeFi addresses on Ethereum (2024)
 - Top 1% hold over 50% of total DeFi TVL

- → ~80,000 wallets qualify
 - **Market Size:**
 - Total TVL: ~\$600B (June 2024)
 - Top-tier TVL: over \$300B
 - **Revenue Potential (Example):**
 - Assuming \$100 annual fee per wallet → \$8M/year
 - Additional income from in-wallet DEX fees and premium privacy plans
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2. Traders Avoiding Sandwich / MEV Attacks

- **Target Segment:** Active DEX users (individual traders)
 - **Stats:**
 - ~2M monthly users across Uniswap, PancakeSwap, 1inch, etc.
 - ~15% are aware of front-running risks → ~300,000 target users
 - **Loss Impact:**
 - Annual MEV-related losses: \$200M–\$400M
 - Avg. loss per trader: ~\$200/year
 - **Revenue Potential:**
 - Assuming 5% adoption of a \$10/month premium plan → 15,000 users = \$150,000/month
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3. Institutional Investors / Enterprises (RWA, B2B Payments)

- **Target Segment:** Companies using blockchain for tokenized assets or B2B payments
 - **Stats:**
 - RWA market size forecast: \$10–16 trillion by 2030 (BCG Report)
 - 10,000+ companies globally exploring or using tokenization
 - Thousands in privacy-sensitive sectors (finance, supply chain, healthcare)
 - **Revenue Potential:**
 - B2B wallet SaaS: \$5,000–\$50,000 per company per year
 - → 100 companies = \$500K–\$5M annual revenue
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4. Tornado Cash / Aztec Connect Migrants

- **Target Segment:**
 - Tornado Cash: Over 250,000 cumulative users
 - Aztec Connect: Over 100,000 users

- Tens of thousands still searching for privacy solutions
- **Behavioral Trends:**
 - Actively researching “next anonymity solution” on GitHub/Twitter
 - Temporarily migrated to Railgun, Noir, zkSync Privacy, etc.
- **Revenue Potential:**
 - Users with high demand for anonymity tend to pay for premium
 - 10,000 users × \$10/month = \$100,000/month

Summary: Combined Impact (Conservative Estimate)

Segment	Potential Users	Annual Revenue Estimate
Large DeFi Users	80,000	Approx. \$8 million
MEV-Resistant Traders	300,000	Approx. \$1.8 million
Enterprise / Institutional	Several thousand	\$500K - \$5 million
Tornado / Aztec Migrants	50,000 - 100,000	Approx. \$1 - 2 million
Total (Conservative)	~500,000 users	Over \$15 million/year

► Use Cases (Situations):

- When users want to conduct **large transfers or investments discreetly**
- When **moving assets or swapping across chains using a DEX**
- When needing a **fast wallet integrated with KYC compliance**

► Product / Solution:

- **Encrypted tokens via eERC standard**
- **Private chain architecture via G-chain**
- **Selective disclosure through KYC triggers** (regulatory compliance)
- **Front-running resistant built-in DEX features**

► Competing Products:

- **Tornado Cash**
- **Aztec**
- **Other privacy wallets such as Railgun**

► Differentiation - Reasons Users Will Choose GhostWallet:

- Not fully anonymous, but “**KYC-triggered anonymity**”, a balanced design
- **All-in-one** configuration including **Cross-Chain DEX**
- **High compatibility with Avalanche’s high-speed network**
- **Sustainable, auditable privacy technology** that’s legally compliant
- **MEV and front-running protection** built in by default

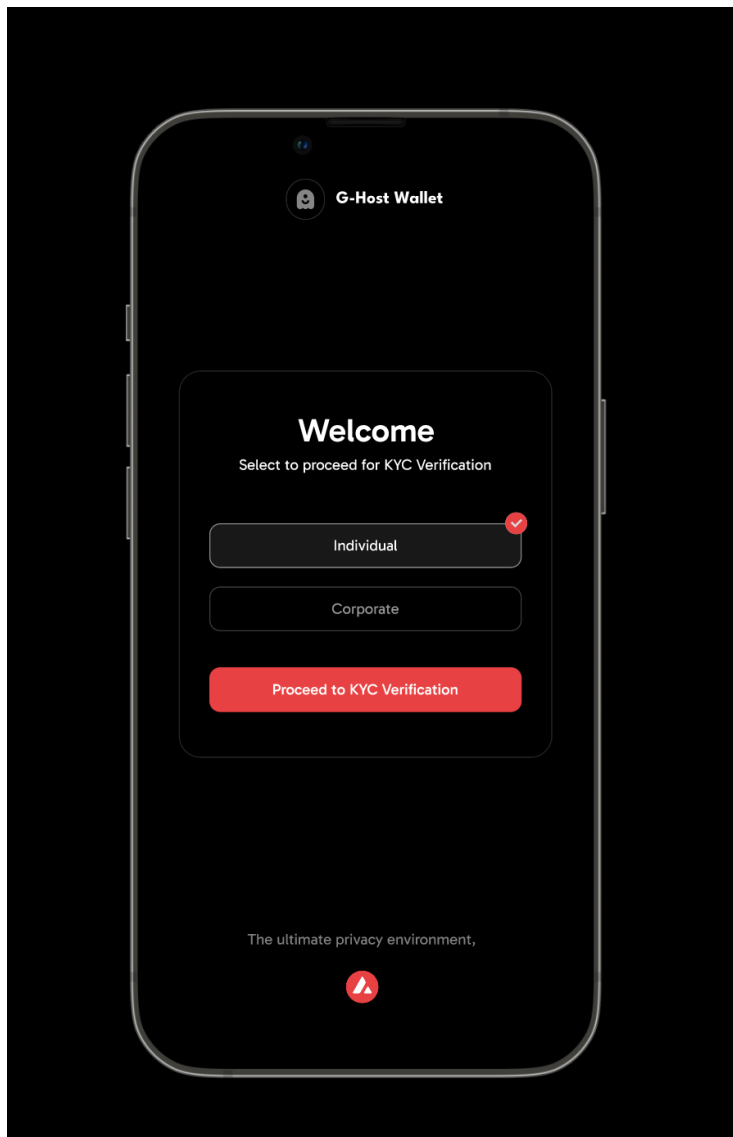
► Therefore...

Users won’t settle for just any overseas privacy wallet — they will *specifically choose GhostWallet* built on Avalanche.

From here onward are the technical and implementation details.

Please include only what’s necessary in your presentation.

GhostWallet: The Potential of a Privacy-First Wallet Built on Avalanche



GhostWallet: A Privacy-Focused Wallet Built on Avalanche

GhostWallet is an innovative wallet designed to address core Web3 challenges—privacy, MEV, and cross-chain DEXs—from a perspective that resonates with Avalanche Codebase mentors. Leveraging Avalanche’s fast and flexible blockchain infrastructure, GhostWallet overcomes the limitations faced by legacy privacy solutions and aims to balance user experience with regulatory compliance.

This report details the issues with existing privacy solutions, the technological uniqueness of GhostWallet, its MEV resistance, its synergy with the Avalanche ecosystem, the benefits of KYC-triggered anonymity, and its long-term potential including token issuance and enterprise integrations.

Challenges with Existing Privacy Solutions and Their Social Impact

To secure privacy on the blockchain, solutions such as Tornado Cash and Aztec emerged. However, while effective, these approaches have faced increasing regulatory and sustainability challenges.

- **Tornado Cash Case:**

Tornado Cash, a leading Ethereum-based mixing service, offered strong anonymity but was abused for money laundering. Since its launch in 2019, it has been linked to over \$7B in illicit flows, including funds laundered by North Korean hacker groups. In August 2022, the U.S. OFAC imposed sanctions, leading to developer arrests and a sharp decline in protocol usage. Although in March 2025 OFAC officially lifted the sanctions following a U.S. appeals court ruling that smart contracts themselves are not “persons,” criminal proceedings against core developers continue. This underscores how fully anonymous privacy protocols remain vulnerable to legal and social backlash.

- **Aztec Case:**

Aztec was a Layer 2 privacy solution for Ethereum, enabling private DeFi via zkSNARKs. Its product, Aztec Connect, was used by over 100,000 users but was sunset in March 2023. While Aztec cited commercial reasons and not regulatory pressure, the shutdown reflected uncertainties around privacy tech. Users had to withdraw funds by March 2024, causing a temporary loss of privacy-focused infrastructure in the ecosystem. The case illustrates that even with solid tech, operational longevity and compliance strategies are critical.

These examples highlight a common dilemma: “**Privacy vs. regulatory compliance.**” Ignoring regulation (like Tornado) risks shutdowns, while staying legal (like Aztec) may still require strategic pivots. A new solution must offer **both privacy and legal usability**, with trust and ease of use. GhostWallet is built precisely to solve this.

GhostWallet’s Technical Uniqueness

(eERC, G-chain, ZK Proofs, DEX Structure)

GhostWallet is a privacy-first wallet that fully utilizes Avalanche’s latest tech stack, integrating encrypted token standards, dedicated chain architecture, ZK proofs, and DEX functionality.

- **Encrypted ERC (eERC) Standard:**

Developed by Avalanche, eERC encrypts token balances and transfer amounts, ensuring confidentiality on-chain. It supports zero-knowledge proofs and homomorphi

c encryption to verify correctness without exposing transaction details. Uniquely, it includes auditability: authorized entities can decrypt under permissioned scenarios, balancing **privacy with compliance**. GhostWallet uses this as its core for private token management.

- **Dedicated G-chain:**

GhostWallet operates on a custom Avalanche subnet—"G-chain"—built using Avalanche. This combines public chain speed and compatibility with private chain confidentiality. eERC tokens on G-chain remain EVM-compatible while adding enhanced privacy. GhostWallet customizes consensus, node configuration, and transaction logic for privacy-centric performance.

- **ZK Proofs for Confidentiality and Validation:**

GhostWallet uses ZK proofs embedded in its Privacy SDK. Users can validate transactions (e.g., sufficient balance, no double spend) without revealing amounts or addresses. Even identity (KYC) can be proven without direct disclosure. This allows secure, private operations with a simple UX and no need for deep cryptographic knowledge.

- **Cross-Chain DEX Structure:**

GhostWallet's built-in DEX enables seamless asset swaps across chains (e.g., eERC tokens on Avalanche ↔ Ethereum assets). G-chain acts as a private bridge hub, encrypting all inter-chain flows. Utilizing Avalanche Warp Messaging, assets move privately across subnets. Users avoid bridge complexity and multi-wallet setups. The DEX also integrates with existing AMMs and bridge aggregators to offer optimal swap rates. GhostWallet stands out as a **multi-chain-ready privacy DEX**, unlike older single-chain models.

Resistance to MEV & Front-Running

GhostWallet counters MEV (Maximal Extractable Value) and front-running attacks using privacy and speed. These attacks exploit mempool visibility—malicious bots scan pending transactions and front-run large trades (e.g., sandwich attacks). On Ethereum, bots like "Jaredfromsubway.eth" made over \$6M this way.

GhostWallet encrypts transaction metadata (recipient, token, amount), making mempool monitoring useless. Even if intercepted, content remains inaccessible without keys. Combined with Avalanche's **instant finality**, which leaves no time window for transaction reordering, this creates a dual defense: **privacy protection and ultra-fast settlement**. Traders get fair prices and avoid slippage losses.

Synergy with Avalanche Ecosystem

GhostWallet is optimized for Avalanche's low-latency, high-throughput blockchain:

- **High Compatibility:**
eERC is EVM-based, enabling GhostWallet to interact with existing Ethereum-based protocols and assets while benefiting from Avalanche's speed and subnets.
- **Low Fees & Scalability:**
Avalanche enables fast (<1s) confirmations with fees < \$1, overcoming common pain points of privacy protocols like high latency and gas costs.
- **Ecosystem Integration:**
Avalanche has committed to privacy (e.g., acquiring EtraPay, launching eERC). GhostWallet fits this strategy and aligns with Avalanche Codebase goals. Its use of Avalanche Cloud, subnets, and Warp Messaging makes it a natural candidate for Codebase mentorship.
- **Cross-Subnet Liquidity Bridge:**
GhostWallet can mediate private liquidity between Avalanche's many subnets (e.g., games, Evergreen for institutions). It may also connect Avalanche's decentralized ID/auth infrastructure with its KYC-triggered privacy.

GhostWallet is, in short, **a privacy wallet only possible because of Avalanche.**

Legal & UX Advantages of KYC-Triggered Anonymity

GhostWallet introduces a **"KYC-triggered anonymity"** model. Users stay fully anonymous during normal use, but identity can be disclosed **only under predefined triggers** (e.g., legal orders, suspicious activity).

- **Legal Benefits:**
Through eERC's audit module, approved authorities can decrypt specific transactions if legally required. This balances user privacy with regulatory needs like AML and sanctions compliance, avoiding the "wild west" issues seen with Tornado Cash.
- **UX Benefits:**
 - **No KYC on entry** = fast onboarding and no friction
 - Default privacy = peace of mind for users (no data exposure)
 - Optional and minimal disclosure = trust through transparency
Even in the event of triggered KYC, the system notifies users and follows clear legal protocols, offering **user-friendly compliance**.

GhostWallet thus delivers **both the peace of privacy and the peace of legal certainty**, setting it apart from all other privacy projects—and standing out in Avalanche Codebase evaluations.

Future Outlook (Token & Enterprise Potential)

GhostWallet is not just technically strong—it also has significant business and expansion potential:

1. Native Token & Ecosystem Growth:

GhostWallet may launch a native token (\$GHOST) to support staking, incentivize liquidity, and govern privacy/audit policies. Token holders could vote on sensitive decisions like audit key managers or KYC thresholds. It may also be listed or used in strategic collaborations. Like other privacy wallet tokens (e.g., \$CWT), this could offer strong investment appeal.

2. Enterprise Use Cases:

GhostWallet can be adopted by institutions and enterprises for confidential B2B payments and tokenized RWA transactions.

- Banks could settle transactions without exposing amounts or clients
 - Supply chains could exchange encrypted pricing data
 - Healthcare/public entities could manage private records on-chain
- Via APIs and SDKs, GhostWallet supports system integration, and enterprises can deploy private GhostChains via AvaCloud. Combined with Avalanche's regulatory frameworks, this creates **trustworthy, compliant blockchain infrastructure**.

3. Roadmap Highlights:

- Expand privacy features (e.g., hide addresses, anonymous messaging)
 - Multi-chain bridge to Ethereum, other L1s
 - Marketing for various segments: DeFi users, traders, NFT holders, institutions
 - Long-term platformization: let developers build privacy-first DApps on GhostWallet
- This vision positions GhostWallet as **not just a wallet, but a privacy infrastructure layer** for Web3.

Conclusion

GhostWallet is a future-proof project across tech, market, and business dimensions. It directly tackles the privacy dilemma, considers real-world regulatory needs, and is designed

for mass adoption.

With strong synergy with Avalanche, GhostWallet is poised to become a **game-changer for Web3 privacy and compliance.**

References:

- Avalanche Blog: Encrypted ERC (eERC) Launch
- Avalanche Docs: eERC & Privacy SDK
- U.S. Treasury Press: Tornado Cash Sanctions
- Venable LLP: Legal Analysis of Sanctions Removal
- Aztec Blog: Aztec Connect Wind-down
- CoinMarketCap: MEV / Front-Running Overview
- Binance Square: eERC Use Cases and Developer Info

Before vs After – Why GhostWallet Is Inevitable	
Current Landscape – Fragmented & Risky	GhostWallet Architecture – Integrated & Private by Design
✗ Tornado Cash • Strong anonymity • Regulatory ban (OFAC sanctioned) • No KYC / No auditability • Service disruption risks	✓ eERC Standard • Encrypted token balances • ZK proof enabled • Optional audit access for compliance • Built-in privacy with compliance control
✗ Aztec Connect • zk-based privacy • Shutdown in 2024 • Limited to Ethereum • Not cross-chain	✓ G-chain • Dedicated Avalanche subriety • Customizable L1 optimized for privacy • Scalable, fast finality
✗ MEV / Sandwich Attacks • Traders lose money • Bots exploit mempool transparency • No built-in protection in standard wallets	✓ Ghost DEX • Front-running resistant • Fully cross-chain • In-wallet asset swaps • No info leaks during bridging
✗ Cross-chain UX Problems • Needs multiple wallets & bridges • Manual operations are error-prone	✓ UX Advantage • All-in-one private wallet • No 3rd-party app required • Seamless onboarding for individuals & enterprises
From fragmented, high-risk privacy hacks... To GhostWallet – a next-gen compliant privacy wallet that Avalanche.	

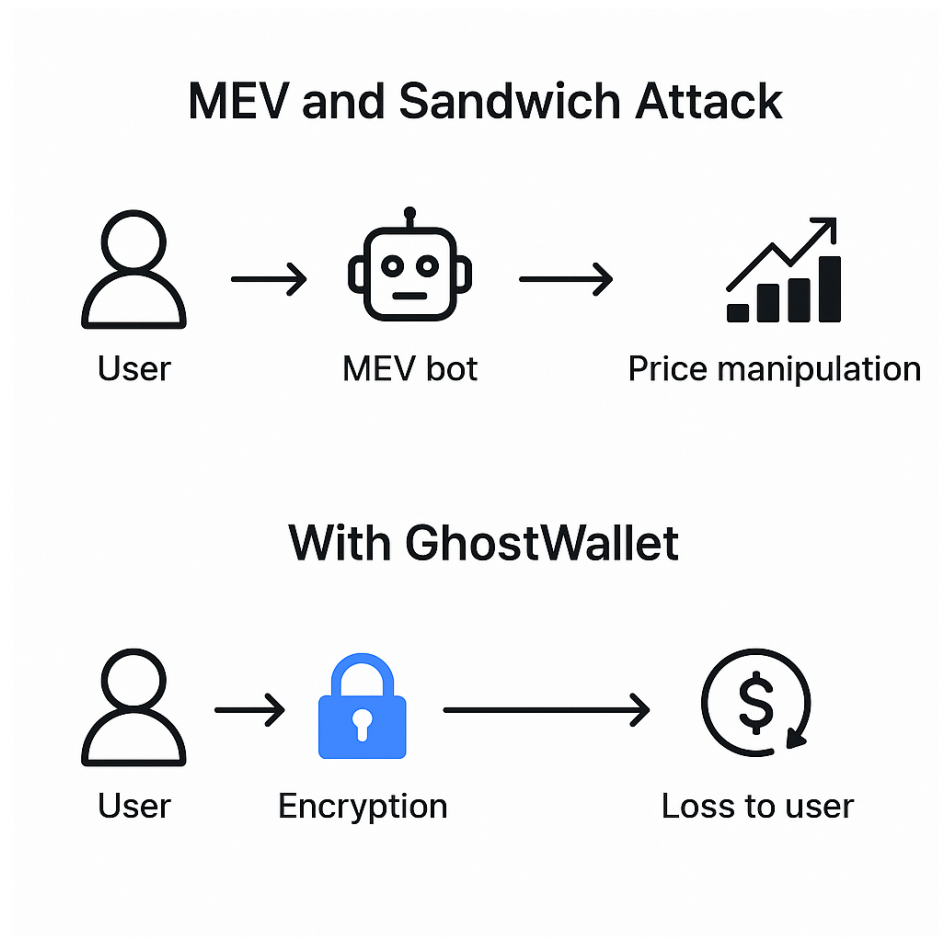
① ✗ Current Landscape vs ✓ GhostWallet Architecture (Before / After Format)

Structure:

- **Left Side:** Current problems
(Tornado / Aztec / MEV attacks / Complex UX)
- **Right Side:** After implementing GhostWallet
(Solved through eERC + G-chain + DEX)

Purpose:

- Clearly communicate *why GhostWallet is necessary* at a glance.



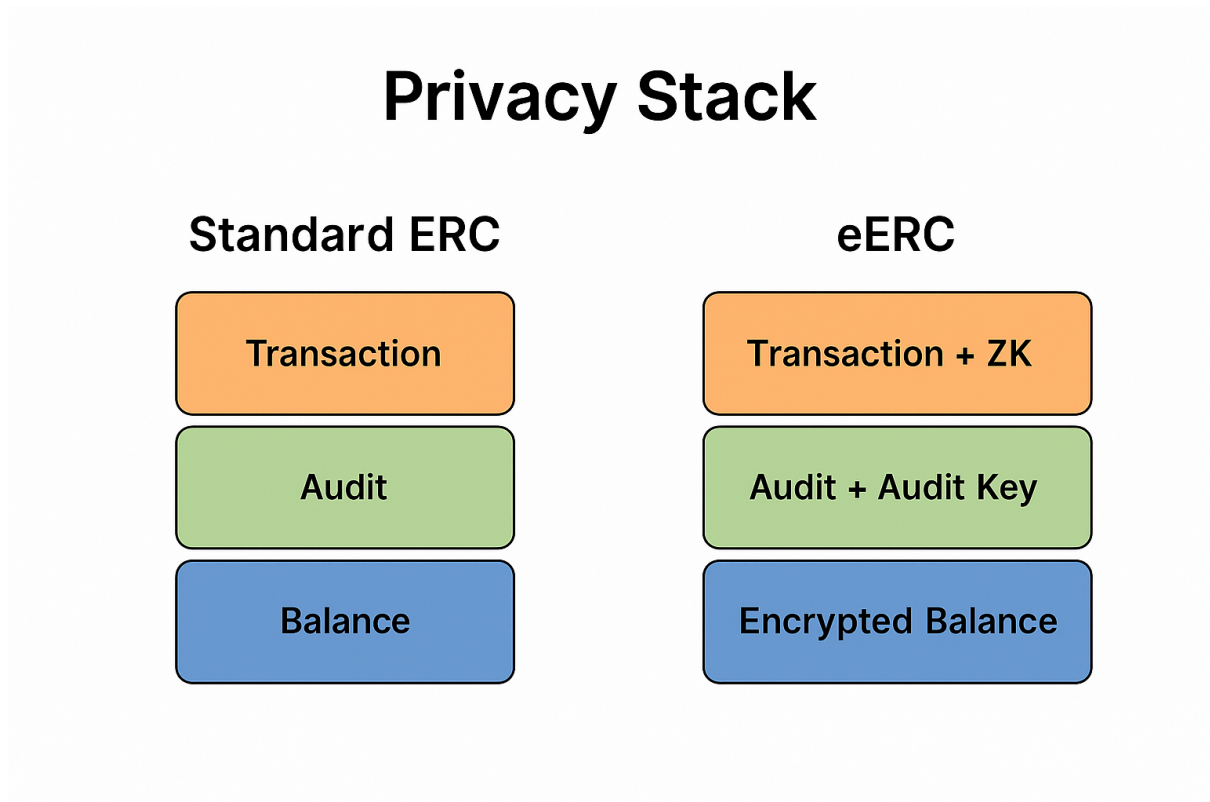
② MEV / Sandwich Attack Diagram (Transaction Sequence)

Structure:

- **Before:**
User → MEV bot → Price manipulation → User loss
- **After GhostWallet implementation:**
Transactions are encrypted → Bots cannot intervene

Purpose:

- Clearly demonstrate a real-world issue and how GhostWallet provides a concrete solution.



③ eERC構造のレイヤー図 (Privacy Stack)

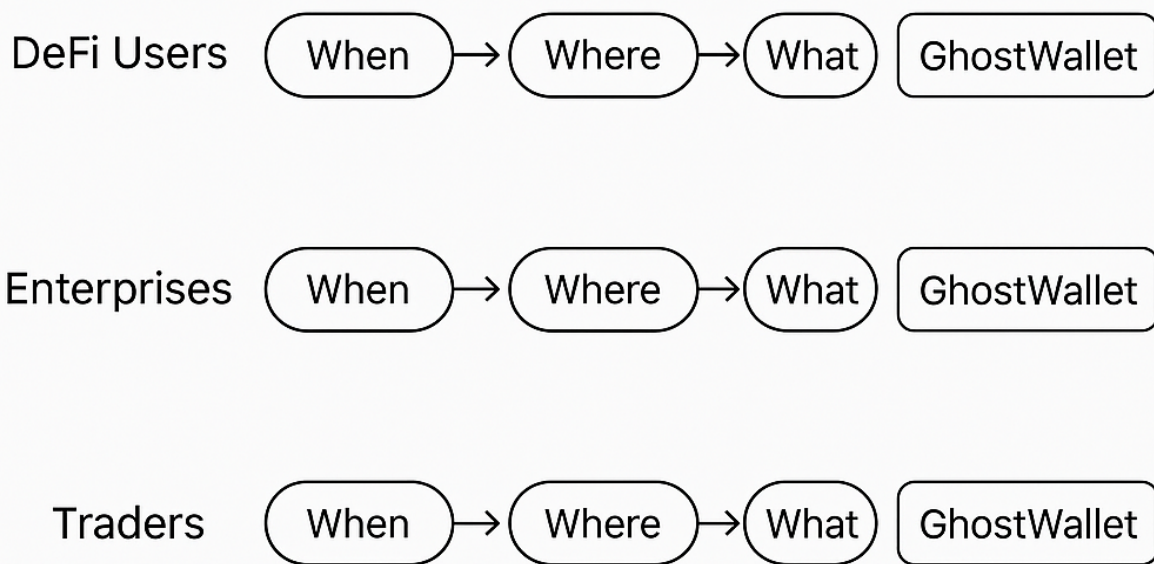
構成 :

- 通常ERC vs eERCをスタック型で比較
- 各層 (ZK, encrypted balance, audit keyなど) を色分け

目的 :

- 技術的独自性と「合法×匿名」の仕組みを伝える

User Journey



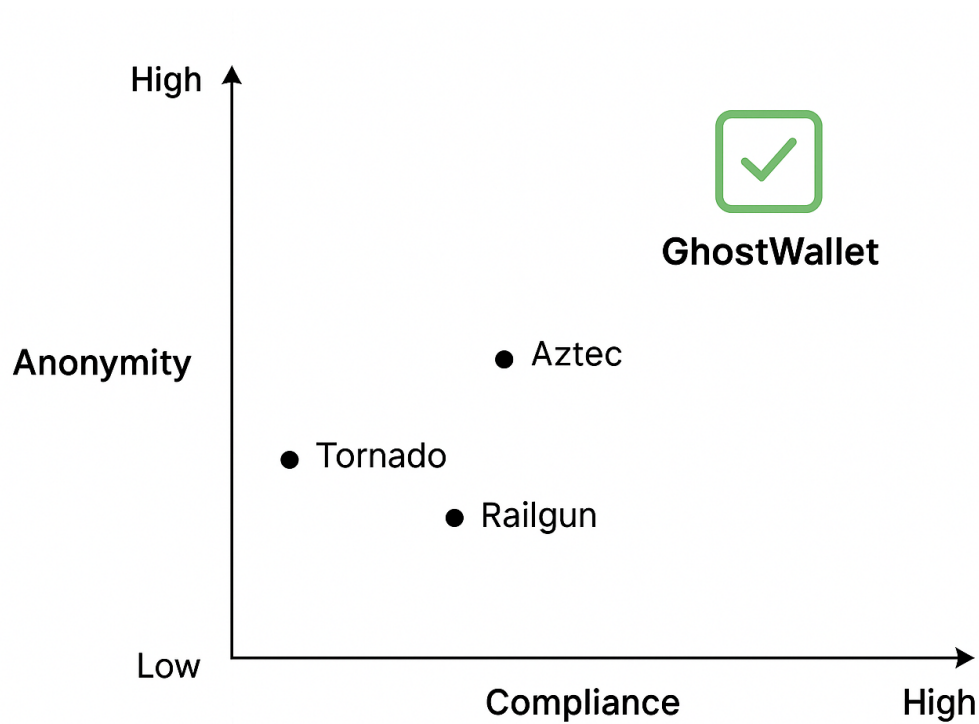
④ User Journey Diagram (Use-Case Flow by Scenario)

Structure:

- Flow of how DeFi users / enterprises / traders use GhostWallet
- Illustrate *when*, *where*, and *for what purpose* it is used

Purpose:

- Help visualize the target users and specific use cases clearly
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⑤ Differentiation Matrix (Competitive Comparison Chart)

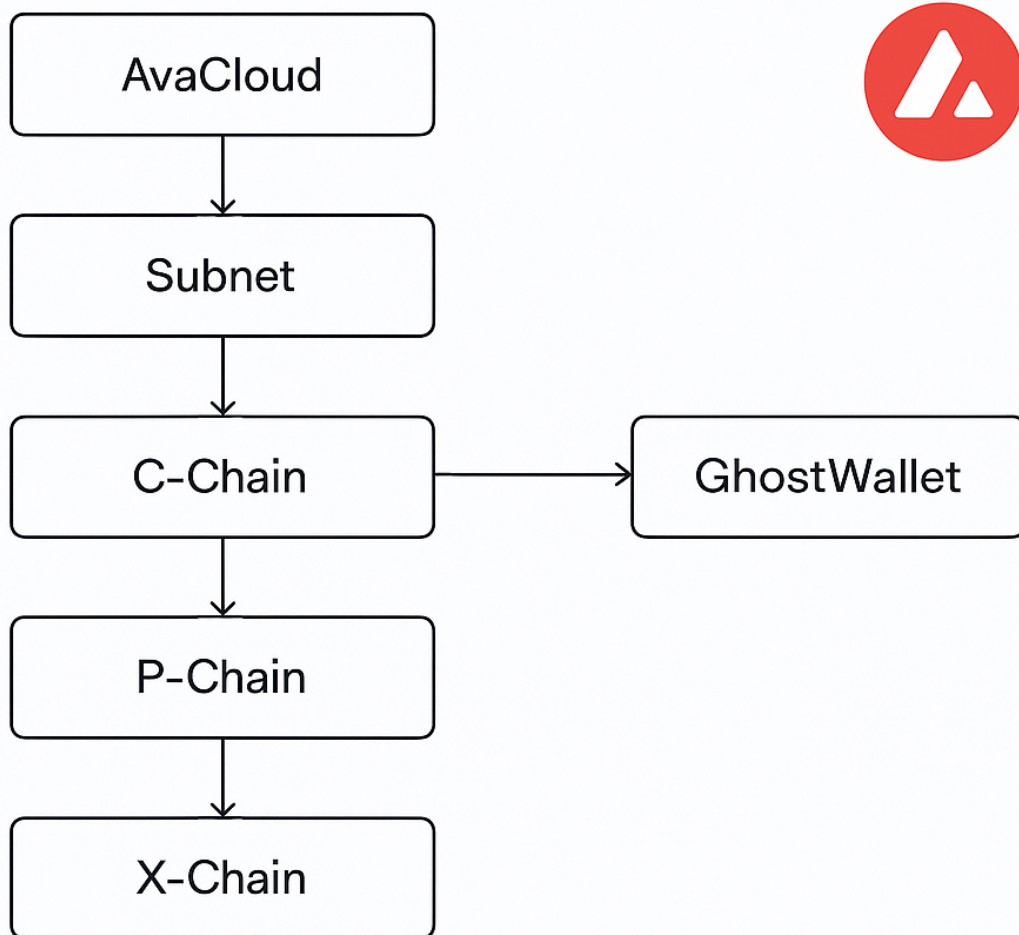
Structure:

- **X-axis:** Compliance readiness
- **Y-axis:** Level of anonymity
- Plot each product (Tornado, Aztec, Railgun, Ghost)

Purpose:

- Visually highlight that **GhostWallet is the only solution offering both legality and anonymity**
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Integration with the Avalanche Stack



⑥ Integration Map with the Avalanche Stack (Technical Architecture Diagram)

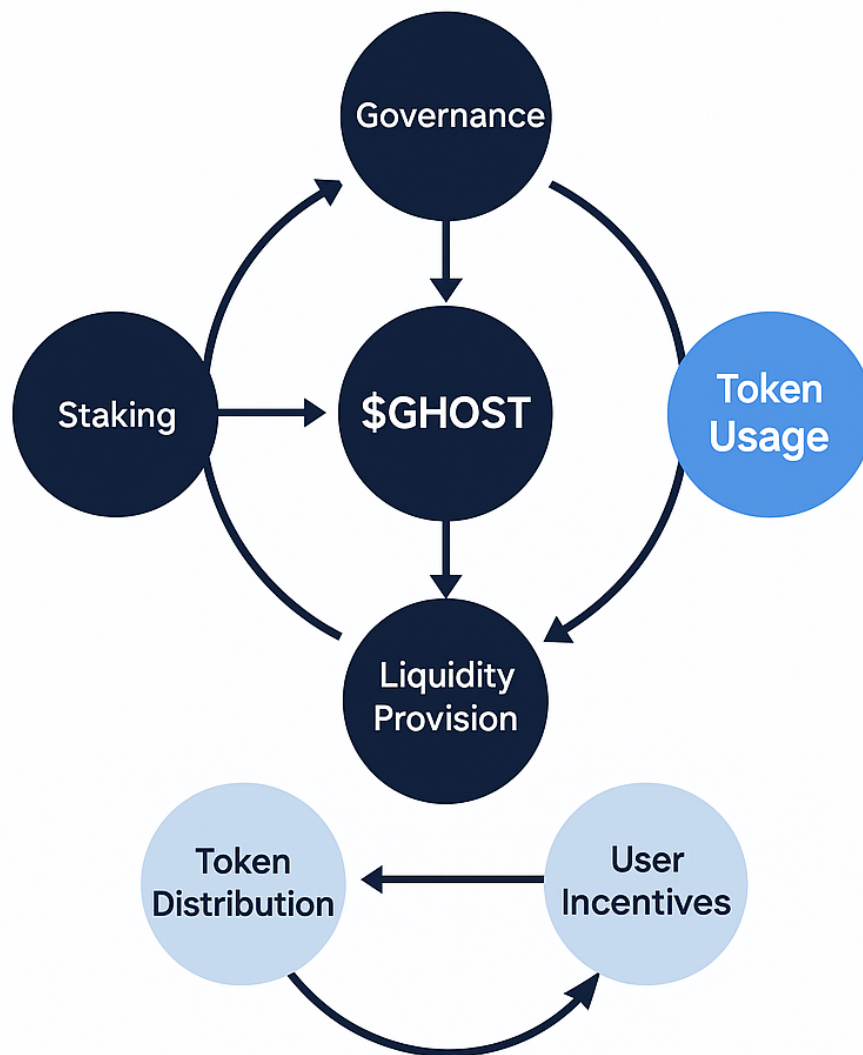
Structure:

- Show connections with AvaCloud / Subnet / Warp / C, P, and X Chains
- Indicate where and how GhostWallet operates within the Avalanche stack

Purpose:

- Clearly justify **why GhostWallet is being built on Avalanche**

Future Ecosystem Development



⑦ Token Economy Model Diagram (Future Ecosystem Expansion)

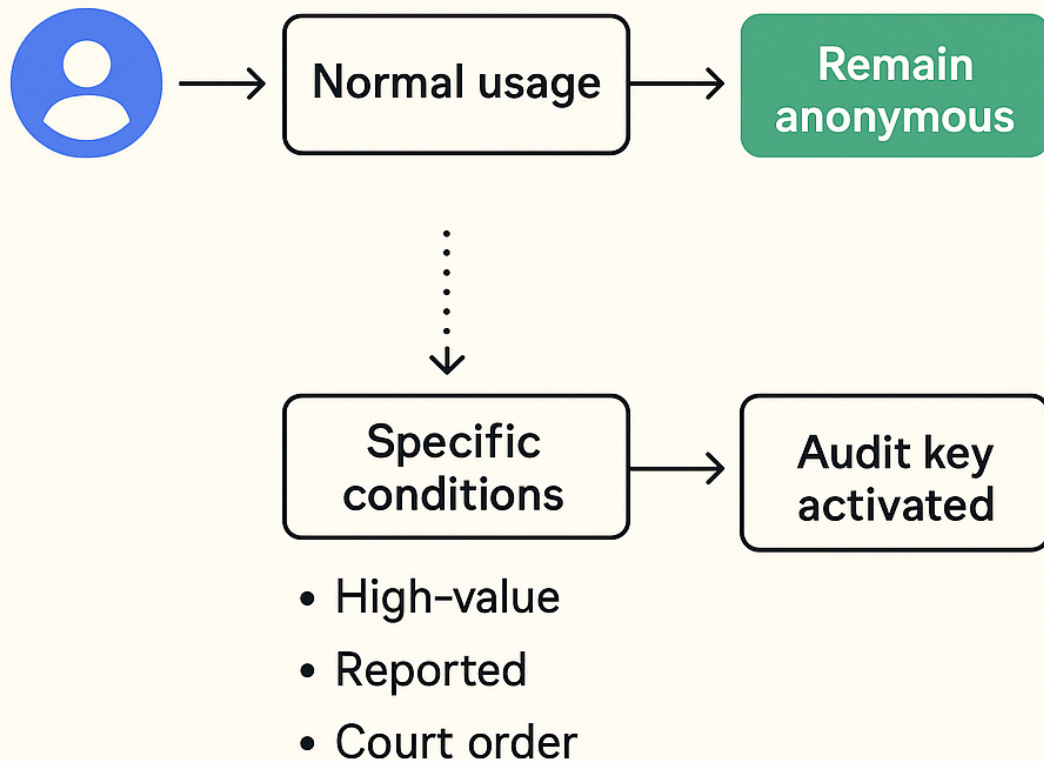
Structure:

- Use cases of \$GHOST token (staking / governance / liquidity provision)
- Flowchart with arrows showing token circulation and user incentive mechanisms

Purpose:

- Demonstrate that GhostWallet is a **sustainable and scalable project**

KYC Trigger-Based UX Flow



Here is the English translation:

⑧ KYC-Triggered UX Flow (Branching Flow Model)

Structure:

- **Normal use:** Fully anonymous is allowed

- **Under specific conditions** (e.g., high-value transactions, flagged activity, court orders), the audit key is activated

Purpose:

- Show that users can use GhostWallet with peace of mind
 - it is **privacy-respecting, not a surveillance system**